

GUJARAT UNIVERSITY

Department of Computer Science

Master of Computer Applications (MCA) - Semester III

Subject: Computer Vision

Practical Assignment - 1

1. Read an image using OpenCV and display its width, height, and number of channels
2. Read an image and save it in another image format
3. Resize an image to 224 x 224 pixels and display the original and resized images
4. Resize an image to 50% of its original dimensions
5. Crop a rectangular region from an image using specified starting and ending coordinates
6. Extract the central region of an image using cropping
7. Convert a color image from BGR/RGB to grayscale
8. Convert an image from BGR/RGB to HSV color space and display the result
9. Convert an image from BGR/RGB to LAB color space and display the result
10. Add salt-and-pepper noise to an image and remove the noise using a median filter
11. Apply a mean filter to an image and observe its effect on image quality
12. Apply Gaussian smoothing using 3x3, 5x5, and 7x7 kernels and compare the results
13. Apply bilateral filtering to an image and observe how it affects noise and edges
14. Apply Gaussian, median, and bilateral filters to the same noisy image and compare their results
15. Sharpen an image using a Laplacian-based sharpening technique
16. Apply unsharp masking to enhance the details and edges of an image
17. Perform contrast stretching on a low-contrast grayscale image
18. Apply histogram equalization and display the image and its histogram before and after processing
19. Apply CLAHE to a low-contrast image and compare the result with standard histogram equalization
20. Normalize an image from the pixel range 0-255 to 0-1
21. Standardize the pixel values of a grayscale image using its mean and standard deviation
22. Apply global thresholding to a grayscale image using a threshold value of 127
23. Apply Otsu's thresholding method to automatically determine a suitable threshold for an image
24. Apply adaptive mean and adaptive Gaussian thresholding to an image and compare the results
25. Convert a grayscale image into a binary image and analyze the effect of different threshold values
26. Apply erosion and dilation to a binary image and display the results

- 27.** Apply morphological opening and closing to a binary image containing small noise and gaps
- 28.** Apply Sobel, Prewitt, Laplacian, and Canny edge detection to the same image and compare the detected edges
- 29.** Detect the edges of an object using Canny edge detection and display the resulting edge image
- 30.** Perform background removal using grayscale conversion, thresholding, and a binary mask, then display the extracted foreground object